

Lecture Log

MA 225

Textbook references correspond to Calculus: Early Transcendentals, 3rd ed., by Briggs, Cochran, Gillett, and Schulz.

Lecture 1: Vectors, Dot Product, and Projection

September 3

§13.1–13.3

Introduces vectors in two and three dimensions, including coordinate notation, magnitude, direction, arithmetic, and the standard basis. Dot product, compute angles between vectors, orthogonality, vector projection.

Lecture 2: Cross products, lines, and planes

September 8

§13.4–13.5

Develops the cross product, both its formula and geometric meaning with applications to area and torque. The lecture then builds vector, parametric, and symmetric equations for lines and point-normal equations for planes.

Lecture 3: Distances and geometry of lines and planes

September 10

§13.5

Constructs planes from geometric data and develops formulas for distances from points to lines and planes. Applications include closest points, skew lines, angles between planes, and tests for coplanarity.

Lecture 4: Quadric surfaces and vector-valued functions

September 15

§13.6; 14.1–14.3

Classify common quadric surfaces using equations, traces, and graphs. The second half introduces vector-valued functions and treats limits, continuity, differentiation, and integration componentwise, leading to velocity, acceleration, speed, and tangent lines.

Lecture 5: Motion, Arc Length, and Multivariable Functions

September 17

§14.3–14.4; 15.1

Apply vector-valued functions to motion and projectile problems, then develops arc length. Introduce arc-length parametrization. Intro to multivariable functions through examples involving domains, graphs, and traces.
